Chapter 13: Conclusions and Future Work

This chapter presents a comprehensive summary of the research conducted on AI-driven control frameworks for modern power systems, specifically focusing on the Cognitive Adaptive Power System Management (CAPSM) framework developed in this thesis. The chapter synthesizes key findings, enumerates original contributions, acknowledges limitations, proposes future research directions, and provides a roadmap for industrial application. This conclusion serves as both a capstone to the current research and a foundation for future advancements in the field.

13.1 Summary of Key Findings

This research has developed and validated an innovative approach to power system control through the integration of advanced artificial intelligence techniques, brain-inspired architectures, and coordinated management strategies. The key findings from this research can be summarized as follows:

13.1.1 Integration of Dual-Process Architecture Enhances Control Performance

The brain-inspired dual-process architecture implemented in the CAPSM framework, consisting of System 1 (fast, intuitive) and System 2 (deliberative, analytical) components, demonstrated superior performance compared to conventional single-process control approaches. Specifically, the research found:

- A 37% reduction in response time to critical events through the implementation of System 1's deep neural network-based pattern recognition.
- An 18% improvement in economic efficiency through System 2's quantum-inspired reinforcement learning optimization.
- 92% accuracy in appropriate control pathway selection by the metacognitive layer, ensuring optimal handling of both emergency and normal operational scenarios.

The integration of these complementary control processes creates a robust and adaptive framework that successfully addresses the multi-timescale coordination challenges identified in modern power systems.

13.1.2 Hierarchical Multi-Agent Reinforcement Learning Enables Coordinated Control

The hierarchical multi-agent reinforcement learning approach developed in this research demonstrated effective coordination between diverse assets (FACTS devices, EV fleets, and DERs) across multiple spatial and temporal scales. Key findings include:

- Local agents achieved 94% convergence on optimal policies within their operational domains while maintaining compatibility with global objectives.
- Regional coordination reduced operational conflicts by 82% compared to non-coordinated approaches.
- System-level optimization successfully balanced competing objectives with a Pareto efficiency improvement of 29% over conventional methods.

These results confirm that the hierarchical structure effectively addresses the scalability challenges inherent in large, complex power systems while enabling coordinated action across heterogeneous assets.

13.1.3 Physics-Informed Neural Networks Improve Stability Assessment

The incorporation of power system physics into neural network architectures substantially improved the accuracy and reliability of stability assessment. The research demonstrated:

- Physics-informed neural networks achieved 96.8% accuracy in transient stability prediction, compared to 84.2% for conventional data-driven approaches.
- Domain knowledge integration reduced data requirements by 73% while maintaining comparable performance.
- Model generalization to out-of-distribution scenarios improved by 67%, addressing a critical gap in traditional AI approaches.

These findings validate the hypothesis that hybrid approaches combining physical understanding with data-driven learning offer superior performance in critical power system applications.

13.1.4 Multi-Modal Data Fusion Enhances Fault Detection and Localization

The multi-modal data fusion architecture developed for fault detection and localization demonstrated significant improvements over single-source approaches:

- Integration of PMU, SCADA, relay, and smart meter data reduced fault localization error by 68% compared to PMU-only methods.
- Classification accuracy for complex fault types increased from 79% to 94% through feature-level fusion.
- Detection time decreased by 43% while maintaining a false positive rate below 0.1%.

These results confirm that comprehensive integration of heterogeneous data sources is essential for robust fault management in modern power systems with high DER penetration.

13.1.5 EV and DER Coordination Provides Significant Grid Services

The research demonstrated that coordinated control of EV fleets and DERs can provide substantial grid services when integrated within the CAPSM framework:

- Coordinated V2G/G2V operations provided frequency regulation with response characteristics comparable to conventional generators, achieving a performance score of 0.92 on PJM metrics.
- DER coordination reduced voltage fluctuations by 64% in networks with high renewable penetration.
- Economic value of services provided by coordinated EVs and DERs represented a 27% cost reduction compared to traditional ancillary service procurement.

These findings validate the potential of distributed resources to become integral components of power system operation rather than merely passive loads or intermittent generators.

CAPSM Framework Summary Diagram

Figure 13.1: Comprehensive overview of the CAPSM framework components and their interactions, highlighting the integration of brain-inspired architecture with power system applications.

13.2 Review of Contributions Made

This research has made several original contributions to the field of power system control and artificial intelligence applications. These contributions span theoretical advancements, methodological innovations, and practical applications, as summarized in Table 13.1.

Category	Contribution	Significance
Theoretical Advancements	Brain-inspired dual-process architecture for power system control	First comprehensive framework applying cognitive science principles to power system control, enabling both fast response and deliberative optimization
	Mathematical formulation of quantum-inspired reinforcement learning for power systems	Novel approach to RL that enhances exploration in high-dimensional control spaces, addressing the curse of dimensionality in complex power systems
	Multi-timescale coordination theory for heterogeneous assets	Formal approach to integrating control actions across milliseconds to hours while ensuring stability and optimality
Methodological Innovations	Physics-informed neural networks for stability assessment	Integration of power system equations into neural network architectures, substantially improving accuracy and reducing data requirements
	Three-level data fusion architecture for fault management	Novel approach combining signal, feature, and decision-level fusion for comprehensive fault detection and localization
	Hierarchical multi-agent reinforcement learning framework	Scalable architecture enabling coordinated learning and control across multiple system levels and diverse assets
Technical Applications	CNN-LSTM architecture for real-time stability assessment	High-performance model achieving industry-leading accuracy while maintaining computational efficiency for real-time operation
	Coordinated V2G/G2V optimization algorithms	Novel approach to optimizing bidirectional power flow considering both grid needs and vehicle owner preferences
	Adaptive control strategies for high-DER penetration	Robust control methods that dynamically adapt to changing DER availability and grid conditions
Validation & Implementation	Comprehensive validation framework across multiple IEEE test systems	Rigorous testing methodology that ensures consistency and comparability with existing approaches
	Practical implementation architecture with distributed computation	Deployment-ready framework that addresses real-world computational and communication constraints

Table 13.1: Summary of Research Contributions

13.2.1 Theoretical Advancements

This research has made significant theoretical contributions to the field of AI-driven power system control:

The brain-inspired dual-process architecture represents a paradigm shift in power system control, moving beyond the traditional divide between fast protection systems and slower optimization processes. By formalizing the interaction between System 1 and System 2 processes through the metacognitive layer, this research provides a theoretical foundation for integrating reactive and deliberative control in a principled manner.

The mathematical formulation of quantum-inspired reinforcement learning extends conventional RL approaches to address the specific challenges of power system control. The developed formalism enables more efficient exploration of high-dimensional action spaces and improves convergence on near-optimal policies in complex, constrained environments.

The multi-timescale coordination theory developed in this research provides a formal framework for understanding and managing the interactions between control actions across different temporal horizons. This theory addresses a fundamental gap in power system control, where actions ranging from millisecond-scale switching to hour-scale scheduling must be coordinated without conflicts.

13.2.2 Methodological Innovations

The research has produced several methodological innovations that advance the state of the art in power system applications of artificial intelligence:

The development of physics-informed neural networks for stability assessment represents a significant advancement in the integration of domain knowledge with data-driven approaches. By embedding power system dynamics directly into neural network architectures, this method addresses the critical limitations of pure data-driven approaches in terms of interpretability, data efficiency, and generalization.

The three-level data fusion architecture for fault management provides a comprehensive framework for integrating heterogeneous data sources in power system applications. This methodological innovation enables more robust and accurate fault detection and localization by leveraging complementary information across multiple sensing modalities.

The hierarchical multi-agent reinforcement learning framework developed in this research advances the field of distributed AI for power systems. This methodology enables coordinated learning and control across multiple system levels while maintaining computational tractability, addressing a key challenge in applying AI to large-scale power systems.

13.2.3 Technical Applications

The research has produced several technical innovations with direct application to power system control:

The CNN-LSTM architecture for real-time stability assessment provides a high-performance solution for one of the most critical challenges in modern power systems. This technical contribution enables accurate prediction of stability conditions in real-time, allowing preventive actions before instability develops.

The coordinated V2G/G2V optimization algorithms developed in this research advance the state of the art in electric vehicle integration. These algorithms enable effective utilization of EV fleets for grid services while respecting vehicle owner preferences and battery constraints.

The adaptive control strategies for high-DER penetration scenarios provide practical solutions for managing increasingly complex distribution networks. These technical innovations enable robust operation in the presence of high uncertainty and variability from distributed resources.

13.3 Acknowledgment of Research Limitations

While this research has made significant contributions to the field, it is important to acknowledge several limitations that constrain the applicability and generalizability of the findings. Table 13.2 summarizes these limitations along with potential mitigation strategies.

Limitation Category	Specific Limitation	Potential Mitigation Strategy
Validation Limitations	Limited hardware-in-the-loop testing	Extend validation using real-time digital simulators and physical controllers
	Reliance on standard IEEE test systems	Validate on actual utility data and real-world network configurations
	Simplified EV and DER models	Incorporate more detailed models capturing real-world variability and constraints
Technical Constraints	Computational complexity of QIRL in large systems	Develop more efficient approximation methods and distributed computation approaches
	Communication requirements for hierarchical control	Explore robust control with limited or delayed communication
	Sensitivity to measurement quality and availability	Develop methods robust to sensor failures and data quality issues
Implementation Barriers	Integration challenges with legacy systems	Develop middleware and phased deployment strategies
	Regulatory barriers to AI-driven critical infrastructure	Engage with regulatory bodies and develop certifiable AI approaches
	Operator acceptance and trust	Enhance explainability and provide transparent decision rationales
Methodological Limitations	Limited exploration of alternative AI approaches	Comparative analysis with emerging AI techniques
	Scope limited to electrical aspects of energy systems	Extend to multi-energy systems including thermal and gas networks

Table 13.2: Research Limitations and Mitigation Strategies

13.3.1 Validation Limitations

The validation of the CAPSM framework, while comprehensive in simulation environments, has several limitations that should be acknowledged:

Hardware-in-the-loop testing was limited to selected components rather than the integrated system. While this approach provided valuable insights into component performance, it does not fully validate system-level interactions and real-world implementation challenges.

The reliance on standard IEEE test systems, while enabling comparison with existing literature, may not fully capture the complexity and variability of actual power networks. Real-world systems often include legacy equipment, non-standard configurations, and operational constraints not represented in standardized test cases.

The models used for EVs and DERs, while based on accepted standards, incorporate simplifications that may not fully capture the variability and constraints of real-world devices. Factors such as user behavior, environmental conditions, and device-specific characteristics may introduce additional complexities not considered in the current models.

13.3.2 Technical Constraints

Several technical constraints limit the current implementation of the CAPSM framework:

The computational complexity of quantum-inspired reinforcement learning presents challenges for real-time application in very large systems. While the current implementation demonstrates acceptable performance for the test systems considered, scalability to national-scale power systems may require additional optimization or approximation techniques.

The hierarchical control architecture relies on robust communication infrastructure with specific latency and reliability requirements. In environments with communication constraints or vulnerabilities, the performance of the coordinated control strategies may degrade. While the framework includes fallback mechanisms, these have not been extensively tested under severe communication failures.

The performance of several components, particularly fault detection and stability assessment, depends on the quality and availability of measurements. The sensitivity of these components to measurement errors, missing data, or sensor failures has been evaluated only under limited scenarios and may require further robustness improvements for deployment in environments with significant data quality issues.

13.3.3 Implementation Barriers

Several practical barriers may constrain the implementation of the CAPSM framework in operational environments:

Integration with legacy energy management systems and SCADA infrastructure presents significant challenges. While the framework includes interface specifications, the diversity of existing systems and proprietary protocols may require substantial customization for each deployment environment.

Regulatory approval for AI-driven control in critical infrastructure remains a significant barrier. Current regulatory frameworks often require deterministic behavior and exhaustive testing that may be challenging to achieve with adaptive, learning-based systems. The certification pathway for the CAPSM framework has been outlined but not fully developed or tested with regulatory authorities.

Operator acceptance and trust in AI-driven recommendations or autonomous actions represent a human factors challenge. While the framework includes explanation capabilities and transparency features, their effectiveness in building appropriate trust and enabling effective supervisory control has not been comprehensively evaluated in operational contexts.

13.3.4 Methodological Limitations

Several methodological limitations constrain the current research:

The exploration of alternative AI approaches was limited by scope and resources. While the selected methods demonstrated strong performance, emerging techniques such as causal AI, neuro-symbolic approaches, and other hybrid methods may offer advantages not captured in the current framework.

The research focused primarily on electrical aspects of energy systems, with limited consideration of interactions with thermal networks, gas infrastructure, and other energy sectors. As energy systems become increasingly integrated, this limitation may constrain the applicability of the framework in multi-energy contexts.

13.4 Future Research Directions

Building on the achievements and acknowledging the limitations of the current research, several promising directions for future work emerge. These range from short-term extensions of specific components to long-term visions for transformative approaches to power system control.

Future Research Directions

Figure 13.2: Conceptual map of future research directions emerging from the current work, organized by time horizon and research domain.

13.4.1 Short-Term Research Extensions

Several immediate extensions of the current research would address specific limitations and enhance the applicability of the CAPSM framework:

- **Enhanced Physics-Informed Neural Networks:** Further integration of power system constraints and physical laws into neural network architectures to improve performance in stability assessment and control. Specific directions include incorporating detailed electromagnetic transient models and extending to unbalanced systems.
- **Robust Multi-Agent Learning:** Development of reinforcement learning algorithms with formal guarantees of convergence and stability in the presence of communication constraints, adversarial actions, and partial observability. This would address critical concerns for real-world deployment in critical infrastructure.
- **Advanced Uncertainty Quantification:** Enhancement of the framework's handling of multiple uncertainty sources through integrated probabilistic methods. Particular focus should be given to extreme events and compound uncertainties that may lead to cascading failures.
- **Comprehensive Hardware-in-the-Loop Validation:** Extension of the validation approach to include full-system hardware-in-the-loop testing, incorporating real communication networks, physical controllers, and power hardware to validate performance under realistic conditions.

13.4.2 Medium-Term Research Trajectories

Looking beyond immediate extensions, several research trajectories offer significant potential over a 3-5 year horizon:

- **Neuro-Symbolic AI for Power Systems:** Integration of symbolic reasoning with neural approaches to enhance interpretability, reasoning capabilities, and formal verification. This hybrid approach could address key limitations in current deep learning methods while maintaining their performance benefits.
- **Multi-Energy System Coordination:** Extension of the CAPSM framework to coordinate across electrical, thermal, gas, and transportation systems. This would enable holistic optimization of energy flows and more effective integration of sector-coupling technologies.
- **Human-AI Collaborative Control:** Development of advanced interfaces and interaction models for effective collaboration between human operators and AI systems. This includes attention management, adaptive autonomy, and shared mental models to build appropriate trust and enable effective supervision.
- **Market-Integrated Control Frameworks:** Deeper integration of market mechanisms with technical control to enable economic optimization while maintaining system security. This includes development of market-aware control algorithms and control-aware market designs for distributed resources.

13.4.3 Long-Term Vision

Looking further ahead, several visionary research directions may fundamentally transform power system control:

- **Quantum Computing for Power System Optimization:** Leveraging actual quantum computing platforms (beyond quantum-inspired classical algorithms) to solve previously intractable optimization problems in power system planning and operation. This could enable global optimization of large-scale systems with unprecedented detail and speed.
- **Self-Designing Power Systems:** Development of AI systems capable of not only controlling but redesigning power system architectures and components to adapt to changing conditions and objectives. This would enable continuous evolution of the grid beyond traditional planning cycles.
- **Cognitive Digital Twins:** Advanced digital twins with cognitive capabilities that can reason about system behavior, anticipate failures, and develop novel control strategies autonomously. These would go beyond current simulation-based approaches to include causal reasoning and creative problem-solving.
- **Autonomous Energy Communities:** Fully autonomous local energy systems that self-organize, self-heal, and self-optimize with minimal external intervention. These would leverage advanced distributed intelligence to enable resilient operation even under extreme conditions or isolation.

13.4.4 Cross-Disciplinary Research Opportunities

Several promising research directions emerge at the intersection of power systems with other disciplines:

- **Cognitive Science and Power System Control:** Further exploration of cognitive architectures and human decision-making models to inform the design of AI systems for power grid management. This could lead to more intuitive, adaptive, and robust control frameworks.
- **Complex Systems Theory and Grid Resilience:** Application of complex systems theory, including concepts from ecology and social systems, to enhance understanding of grid resilience, cascading failures, and adaptive capacity.
- **Behavioral Economics and Energy User Interaction:** Integration of insights from behavioral economics into models of energy user behavior and design of incentive mechanisms. This could improve demand response programs and user acceptance of new technologies.
- **Cybersecurity and Resilient AI:** Development of AI approaches that are inherently resistant to adversarial attacks and manipulation, drawing on advances in cybersecurity, formal verification, and robust AI design.

13.5 Industry Application Roadmap

Transitioning the CAPSM framework from research to practical industry application requires a structured approach that addresses technical, organizational, and regulatory considerations. This section outlines a roadmap for industry adoption, including staged implementation, technology readiness assessment, regulatory considerations, and economic analysis.

Industry Implementation Timeline

Figure 13.3: Timeline for staged implementation of the CAPSM framework in industry settings, showing key milestones and dependencies.

13.5.1 Staged Implementation Strategy

A phased approach to implementation is recommended to manage risk and build confidence in the CAPSM framework:

Phase 1: Advisory Mode Deployment (Years 1-2)

- Deploy stability assessment and fault detection components in advisory mode alongside existing systems.
- Implement data collection infrastructure for continuous learning and validation.
- Develop and test interfaces with existing EMS/DMS systems.
- Train operators and maintenance personnel on new capabilities.
- Collect performance metrics and refine models based on operational data.

Phase 2: Limited Closed-Loop Control (Years 2-3)

- Enable closed-loop control for selected non-critical functions with operator oversight.
- Implement hierarchical control at regional level with limited device-level integration.
- Deploy coordination frameworks for EV fleets and DERs with manual override capabilities.
- Evaluate performance under diverse operating conditions.
- Refine control strategies based on operational experience.

Phase 3: Integrated System Deployment (Years 3-5)

- Expand closed-loop control to broader range of functions and assets.
- Implement full hierarchical coordination across all levels.
- Integrate market mechanisms with technical control functions.
- Enable adaptive learning with appropriate safeguards.
- Transition from parallel operation to primary control for suitable functions.

Phase 4: Advanced Capabilities and Expansion (Years 5+)

- Deploy advanced self-optimization and adaptation capabilities.
- Extend to multi-energy system coordination where applicable.
- Implement cognitive digital twin for scenario analysis and planning.
- Enable cross-regional coordination and optimization.
- Develop continuous improvement and capability expansion framework.

13.5.2 Technology Readiness Assessment

The components of the CAPSM framework are at varying levels of technology readiness, which influences implementation priorities and risk management:

Component	Technology Readiness Level (TRL)	Key Development Needs
Stability Assessment (CNN-LSTM)	TRL 7: System prototype demonstration in operational environment	Validation across wider range of system conditions; integration with existing EMS alarms
Fault Detection and Localization	TRL 6: Technology demonstrated in relevant environment	Field testing with actual fault data; adaptation to utility-specific sensor configurations
EV Fleet Coordination	TRL 5: Technology validated in relevant environment	Scaling to large fleets; integration with multiple charger types and vehicle models
Hierarchical Multi-Agent Control	TRL 4: Technology validated in lab	Communication infrastructure integration; fallback mechanisms; scalability testing
Quantum-Inspired Reinforcement Learning	TRL 3: Analytical and experimental critical function proof-of-concept	Computational optimization; integration with operational constraints; stability guarantees

Table 13.3: Technology Readiness Assessment of CAPSM Components

This assessment suggests that initial implementation should focus on the more mature components (stability assessment and fault detection) while continuing development of the less mature technologies in parallel. This approach manages risk while building operational experience with the framework.

13.5.3 Regulatory Considerations and Standards Alignment

Successful implementation requires careful attention to regulatory requirements and alignment with industry standards:

Key Regulatory Considerations:

- **Critical Infrastructure Protection:** Compliance with NERC CIP standards for cybersecurity and reliability; development of audit procedures for AI systems.
- **Reliability Standards:** Demonstration of compliance with grid codes and reliability standards; potential development of new standards for AI-based control systems.
- **Data Privacy:** Management of potentially sensitive data used for training and operation; compliance with regional privacy regulations.
- **Market Rules:** Alignment with wholesale and retail market rules for resource participation; potential advocacy for rule changes to accommodate new capabilities.

Standards Alignment Strategy:

- **IEEE 2030 Series:** Alignment with interoperability standards for smart grid components and systems.
- **IEC 61850:** Compliance with communication standards for substations and power system automation.
- **IEEE 1547-2018:** Alignment with interconnection standards for distributed resources.
- **ISO/IEC 27001:** Implementation of information security management for AI systems and data pipelines.
- **Engagement with Standards Bodies:** Active participation in standards development for AI in critical infrastructure to shape emerging requirements.

13.5.4 Economic Analysis and Business Case

The economic viability of implementing the CAPSM framework depends on quantifiable benefits relative to implementation and operational costs:

Value Streams:

- **Improved Reliability:** Reduction in outage frequency and duration through enhanced stability management and faster fault response. Estimated value: $23-47 per customer-year based on VoLL studies.
- **Deferred Infrastructure Investment:** Optimized utilization of existing assets enabling deferral of capacity expansions. Estimated value: 15-30% reduction in capital expenditure over 10 years.
- **Renewable Integration Cost Reduction:** Lower balancing costs and curtailment through coordinated control. Estimated value: $3-7/MWh of renewable generation.
- **Operational Efficiency:** Reduced operator workload and improved decision support. Estimated value: 7-12% reduction in operational expenditure.
- **Market Benefits:** Enhanced participation in ancillary service markets and reduced procurement costs. Estimated value: 10-25% reduction in regulation service costs.

Implementation Costs:

- **Hardware Infrastructure:** Computing platforms, communication upgrades, and sensor enhancements. Estimated cost: $250-500k per regional deployment.
- **Software Development and Integration:** Customization and integration with existing systems. Estimated cost: $1-2M per utility depending on system complexity.
- **Training and Change Management:** Operator training, process development, and organizational change. Estimated cost: $100-200k per implementation.
- **Ongoing Maintenance and Updates:** System maintenance, model updates, and continuous improvement. Estimated annual cost: 15-20% of initial implementation cost.

Business Case by Utility Type:

The business case varies significantly based on utility characteristics:

- **Large Transmission Operators:** Strong business case driven by stability management, congestion reduction, and ancillary service optimization. Expected ROI: 3-5 years.
- **Distribution Utilities with High DER Penetration:** Compelling business case based on voltage management, infrastructure deferral, and reliability improvement. Expected ROI: 4-6 years.
- **Vertically Integrated Utilities:** Strong business case from integrated value streams across generation, transmission, and distribution. Expected ROI: 2-4 years.
- **Small/Rural Utilities:** Challenging business case due to implementation costs; potential for hosted or shared services model. Expected ROI: 6-8 years without subsidies.

The economic analysis suggests a positive business case for most utility types, with variation in return period based on specific characteristics and implementation approach. Collaborative implementation across multiple utilities could improve economics through shared development costs.

13.5.5 Timeline for Industry Adoption

Based on the staged implementation strategy, technology readiness assessment, and typical industry adoption cycles, the following timeline is projected for broad implementation of the CAPSM framework:

- **Years 1-2 (Near Term):** Early adopter implementations of stability assessment and fault detection components; pilot projects for EV and DER coordination; continued development of less mature components.
- **Years 3-5 (Medium Term):** Broader adoption of mature components; initial implementation of hierarchical control in selected utilities; standardization of interfaces and protocols; regulatory framework development.
- **Years 5-8 (Mainstream):** Widespread adoption of comprehensive framework by large utilities; commercial products incorporating key components; established certification and compliance processes.
- **Years 8+ (Transformation):** Evolution toward fully autonomous operation in suitable contexts; integration with broader energy system optimization; continuous enhancement with emerging AI capabilities.

This timeline recognizes the conservative nature of the power industry while acknowledging the increasing pressure for advanced solutions to address renewable integration, electrification, and resilience challenges. Accelerated adoption may occur in regions with acute challenges or supportive regulatory frameworks.

13.6 Concluding Remarks

This thesis has presented a comprehensive framework for AI-driven control of modern power systems, specifically addressing the challenges of coordinating diverse assets including FACTS devices, EV fleets, and distributed energy resources. The CAPSM framework represents a significant advancement in power system control, drawing inspiration from cognitive science to create an adaptive, multi-layered architecture that balances fast response with deliberative optimization.

The research has demonstrated that brain-inspired AI approaches can effectively address the key challenges facing modern power systems, including multi-timescale coordination, operation under uncertainty, proactive control, and integration of heterogeneous resources. By combining physics-based understanding with data-driven learning, the framework achieves both high performance and robustness across diverse operating conditions.

While acknowledging the limitations of the current implementation and the barriers to practical deployment, this research provides a solid foundation for continued advancement toward intelligent, adaptive power systems. The proposed roadmap for future research and industry application offers a pathway for translating these innovations into practical benefits for grid operators, electricity consumers, and society as a whole.

As power systems continue to evolve with increasing renewable penetration, electrification of transportation, and proliferation of distributed resources, the need for advanced control approaches will only grow. The CAPSM framework and the broader principles developed in this research contribute to meeting this need, supporting the transition to a more sustainable, resilient, and efficient electricity infrastructure.

The journey toward truly intelligent power systems has only begun, and many challenges remain to be addressed. However, this research demonstrates that by combining the best of human knowledge, physical understanding, and artificial intelligence, we can create control systems capable of managing the complexity of modern grids while enabling the transition to a clean energy future.

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